

Software Engineering Lab

Lab Assignment 3

Understand Jira through YouTube and create use cases, sprints for given project

Agile Backlog Creation & Sprint Simulation in Jira

Hospital Bed & ICU Allocation Dashboard

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1. Epics and User Stories Backlog Summary

HBICU-1 — Patient Triage & Bed Management

FR-001 and FR-002: vitals, severity-based ER ordering, occupancy and bed states.

HBICU-2 — ICU Transfer & Critical Case Management

FR-003 and the FR-004 handoff: scored ICU requests and critical-case escalation.

HBICU-3 — Ambulance Transfers & Audit Accountability

FR-004 and FR-005: transfer decisions, ambulance dispatch and traceable events.

Story	Summary	Epic	Priority	Story Points	Sprint
HBICU-4	Enter and Validate Patient Vitals	1	High	3	1
HBICU-5	Compute Triage Severity and Rank ER Waitlist	1	High	8	1
HBICU-6	View Real-Time Ward and ICU Occupancy	1	High	5	1
HBICU-7	Manage Bed Status	1	High	3	1
HBICU-8	Request ICU Transfer Using Triage Score	2	High	5	1
HBICU-9	Escalate Critical Cases When ICU Is Full	2	High	5	2
HBICU-10	Approve or Reject Inter-Facility Transfer	3	High	5	2
HBICU-11	Dispatch Ambulance for Approved Transfer	3	High	8	2
HBICU-12	Record Bed and Transfer Audit Events	3	High	5	2
HBICU-13	Review Audit Logs and Generate Audit Report	3	Medium	3	2

Estimation and ordering: Fibonacci estimates use 3 for bounded input/status/reporting, 5 for stateful workflows, and 8 for dynamic prioritization or integration uncertainty. Sprint 1 establishes core dependencies (24 points); Sprint 2 adds escalation, dispatch and audit workflows (26 points). Reporting is Medium priority.

Story/use-case detail: Jira descriptions contain the role–goal–benefit story, requirement mapping, preconditions, main and alternative flows, acceptance criteria and estimate rationale. Use cases are embedded in their corresponding stories.

2. Jira Backlog with Epics and User Stories

The screenshot displays the Jira Backlog interface for a project named "Hospital Bed & ICU Allocation Dashboard". The main view is the "HBICU board" in "Backlog" mode. The interface includes a top navigation bar with a search bar, a "Create" button, and a "See plans" button. Below the navigation bar, there are tabs for "Summary", "Timeline", "Active sprints", "Backlog", "Calendar", "List", "Forms", "Development", and "Docs".

The "Backlog" view shows two sprints, "HBICU Sprint 1" and "HBICU Sprint 2", each with 5 work items. The left sidebar shows the "Epic" list with three epics: "Patient Triage & Bed Management", "ICU Transfer & Critical Case Management", and "Ambulance Transfers & Audit Accountability".

HBICU Sprint 1 (5 work items, Estimate: 24)

Story ID	Story Description	Parent Epic	Status	Estimate	Priority	Assignee
HBICU-4	Enter and Validate Patient Vitals	Patient Triage & B...	To Do	3	High	AB
HBICU-5	Compute Triage Severity and Rank ER Waitlist	Patient Triage & B...	To Do	8	High	AB
HBICU-6	View Real-Time Ward and ICU Occupancy	Patient Triage & B...	To Do	5	High	AB
HBICU-7	Manage Bed Status	Patient Triage & B...	To Do	3	High	AB
HBICU-8	Request ICU Transfer Using Triage Score	ICU Transfer & Cri...	To Do	5	High	AB

HBICU Sprint 2 (5 work items, Estimate: 26)

Story ID	Story Description	Parent Epic	Status	Estimate	Priority	Assignee
HBICU-9	Escalate Critical Cases When ICU Is Full	ICU Transfer & Cri...	To Do	5	High	AB
HBICU-10	Approve or Reject Inter-Facility Transfer	Ambulance Transf...	To Do	5	High	AB
HBICU-11	Dispatch Ambulance for Approved Transfer	Ambulance Transf...	To Do	8	High	AB
HBICU-12	Record Bed and Transfer Audit Events	Ambulance Transf...	To Do	5	High	AB
HBICU-13	Review Audit Logs and Generate Audit Report	Ambulance Transf...	To Do	3	Medium	AB

Planning snapshot: all 10 stories, 3 parent epics, priorities and points; Sprint 1 = 24 points, Sprint 2 = 26 points.

3. Story Point Assignments

The screenshot shows a Jira board for 'HBICU board'. The board is divided into two sprints: 'HBICU Sprint 1' and 'HBICU Sprint 2'. Each sprint contains a list of tasks (stories) with their respective story point estimates. The left sidebar shows the epic categories: 'Patient Triage & Bed Management', 'ICU Transfer & Critical Case Management', and 'Ambulance Transfers & Audit Accountability'.

HBICU Sprint 1 (5 work items) | Estimate: 24

Task	Assignee	Priority	Story Points
HBICU-4 Enter and Validate Patient Vitals	Patient Triage & B...	To Do	3
HBICU-5 Compute Triage Severity and Rank ER Waitlist	Patient Triage & B...	To Do	8
HBICU-6 View Real-Time Ward and ICU Occupancy	Patient Triage & B...	To Do	5
HBICU-7 Manage Bed Status	Patient Triage & B...	To Do	3
HBICU-8 Request ICU Transfer Using Triage Score	ICU Transfer & Cri...	To Do	5

HBICU Sprint 2 (5 work items) | Estimate: 26

Task	Assignee	Priority	Story Points
HBICU-9 Escalate Critical Cases When ICU Is Full	ICU Transfer & Cri...	To Do	5
HBICU-10 Approve or Reject Inter-Facility Transfer	Ambulance Transf...	To Do	5
HBICU-11 Dispatch Ambulance for Approved Transfer	Ambulance Transf...	To Do	8
HBICU-12 Record Bed and Transfer Audit Events	Ambulance Transf...	To Do	5
HBICU-13 Review Audit Logs and Generate Audit Report	Ambulance Transf...	To Do	3

Story points are the grey numeric badges beside each story: Sprint 1 totals 24 points; Sprint 2 totals 26 points.

4. Sprint Board (Active Sprint View)

The screenshot shows a Jira Sprint Board for the 'HBICU board'. The board is organized into three columns: TO DO, IN PROGRESS, and DONE. Each column contains story cards with titles, labels, checkboxes, point values, and assignees.

TO DO (1)

- Request ICU Transfer Using Triage Score
ICU Transfer & Critical Case Management
HBICU-8 (5 points, assigned to AB)

IN PROGRESS (2)

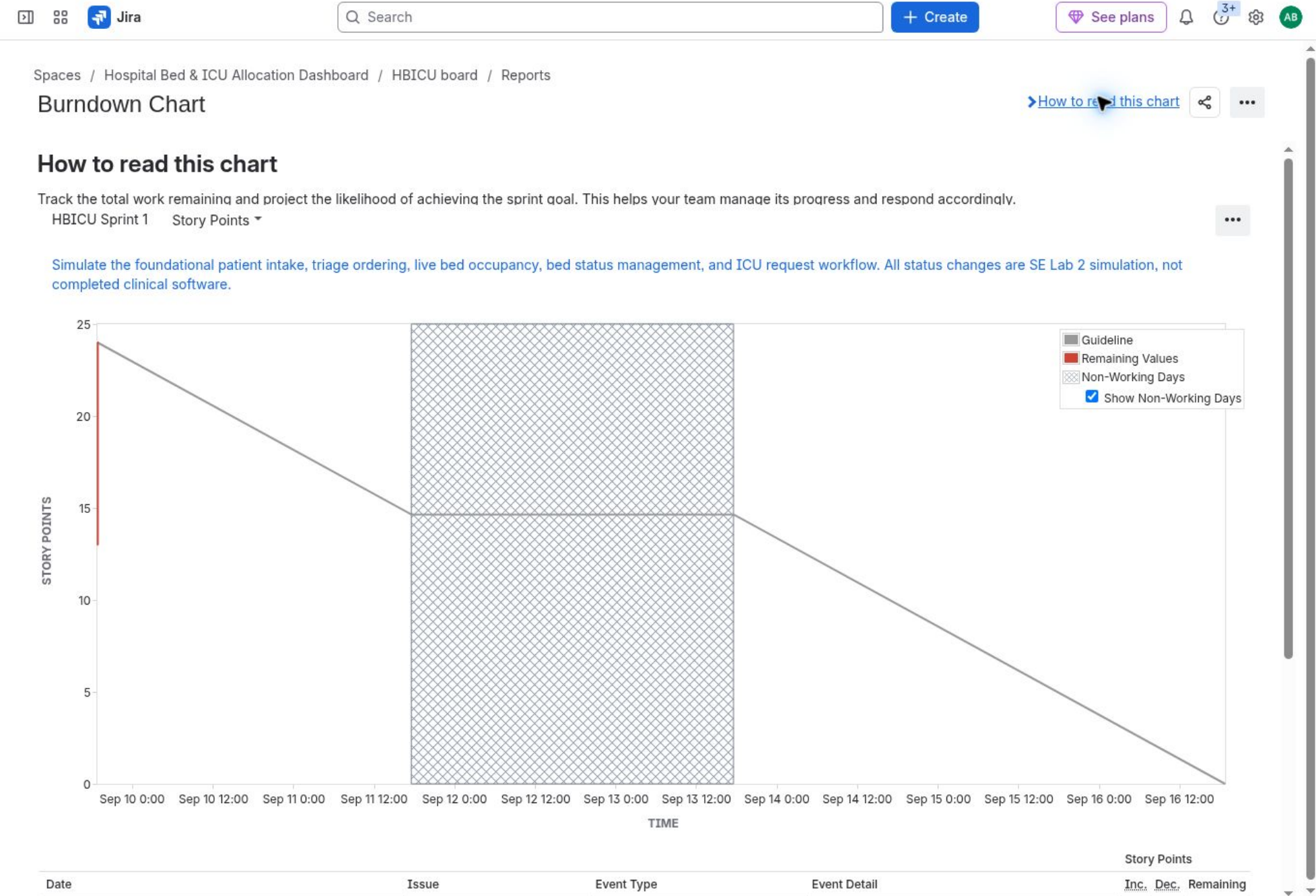
- View Real-Time Ward and ICU Occupancy
Patient Triage & Bed Management
HBICU-6 (5 points, assigned to AB)
- Manage Bed Status
Patient Triage & Bed Management
HBICU-7 (3 points, assigned to AB)

DONE (2)

- Enter and Validate Patient Vitals
Patient Triage & Bed Management
HBICU-4 (3 points, assigned to AB)
- Compute Triage Severity and Rank ER Waitlist
Patient Triage & Bed Management
HBICU-5 (8 points, assigned to AB)

Sprint 1 interim snapshot: To Do = 5 points; In Progress = 8 points; Done = 11 points. All were later completed.

5. Burndown Chart



Actual Jira chart during Sprint 1: 24 initial points decreased to 13; the guideline spans the planned one-week period.

6. Reflection Questions and Answers

1. Did your estimations reflect the actual effort?

The 3-, 5- and 8-point estimates consistently express relative complexity. This was a status simulation without implementation or measured work effort, so actual estimation accuracy cannot be established.

2. Was your backlog well-prioritized?

Core vitals, triage, bed visibility and ICU requests were ranked first. Escalation, approval and dispatch followed their dependencies. Audit event recording remained High priority, while the review/report story was Medium.

3. How did your simulated sprint align with your plan?

Sprint 1 completed its five planned stories (24 points), and Sprint 2 completed its five stories (26 points). Each story moved through To Do, In Progress and Done. The interim Sprint 1 board captured two Done, two In Progress and one To Do story.

4. What insights did the burndown chart give about your team's capacity?

Sprint 1 fell from 24 to 13 points at the interim checkpoint and then to zero. Sprint 2 fell from 26 to zero. These results demonstrate tracking mechanics but do not by themselves establish sustained team velocity.

7. Outcome

The company-managed Scrum workspace contains three functional epics and ten properly formatted, estimated and prioritized stories. Both sprints were configured for a one-week duration and completed in Jira, with zero remaining points. The attached screenshots document planning, active progress, burndown and final completion for instructor inspection.

Reading the evidence: The backlog and active-board images preserve earlier stages, and the burndown chart shows the final state. Times shown are as displayed by Jira.

Submission materials: This PDF. The live workspace remains available for the required instructor demonstration.